

A real-world pharmacovigilance study of nusinersen based on the FAERS database

Weiling Guo

Medical Department, Guangzhou Chest Hospital

Yi Fang

Department of Nosocomial Infection, Guangzhou Chest Hospital

Aiguang Xu

Medical Department, Guangzhou Chest Hospital

Lijuan Wu

Department of General Internal Medicine, Guangzhou Chest Hospital

Yonghong Zou

Outpatient Department, Guangzhou Chest Hospital

Liang Yan

yanl8@mail13.sysu.edu.cn

Department of Nosocomial Infection, Guangzhou Chest Hospital

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Abstract

Objective: This study aimed to evaluate and analyse nusinersen adverse events (AEs) in the FDA Adverse Event Reporting System (FAERS) to provide comprehensive safety data on nusinersen in real-world.

Methods: Reports on nusinersen were extracted from the FAERS database spanning from January 2017 to December 2023. Disproportionality analysis were performed using reporting odds ratio (ROR), proportional reporting ratio (PRR), Bayesian confidence propagation neural network (BCPNN) and multi-item gamma Poisson shrinker (MGPS), to find out the AEs for nusinersen. Additionally, we described the time-to-onset (TTO) of AEs.

Results: The study retained the 227 preferred terms (PTs) identified by all four algorithms. Important AEs such as traumatic lumbar puncture, neuromuscular scoliosis, cerebrospinal fluid (CSF) pressure increased, neurological procedural complication, mechanical ventilation complication, cerebrospinal fluid leakage, CSF protein increased, procedural anxiety, arachnoid cyst, chronic respiratory failure and sudden infant death syndrome. The median TTO for nusinersen AEs was 119.5 days (interquartile range [IQR] 14–492 days). More than 32.01% of AEs occurred after one year.

Conclusion: Our signal detection of nusinersen reported some potential signals of uncommon AEs, which could aid in risk identification and clinical monitoring.

1 Introduction

Spinal muscular atrophy (SMA) is a disease caused by insufficient levels of survival motor neuron protein, leading to the loss of motor neurons in the anterior horn of the spinal cord^[1]. It affects approximately 1 in 10,000 live births and is characterized by muscle weakness, atrophy, and fatigue^[2]. SMA is one of the most fatal genetic disease in children under 2 years old, with respiratory failure being the primary cause of death^[3]. Nusinersen, an antisense oligonucleotide, was approved as the first therapy for SMA in December 2016. Due to its inability to cross the blood-brain barrier, it needs to be repeatedly administered intrathecally over time. The recommended dosing regimen consists of four loading doses, with the first three given at 14-day intervals and the fourth dose administered 30 days after the third dose. Following this, a maintenance dose is given once every four months^[3, 4]. The most frequently reported AEs include pyrexia, scoliosis, upper respiratory tract infection, nasopharyngitis, vomiting, headache, and constipation. However, these AEs may not be directly related to nusinersen but are closely associated with the progression of SMA^[4–6].

Although nusinersen is generally considered tolerable and safe^[7–10], there are concerns regarding the small sample sizes in experimental studies. The early termination of pivotal clinical trials due to ethical reasons led to shortened evaluation periods and data loss^[11]. Additionally, the repeated intrathecal administration of nusinersen raises concerns about potential tissue damage and infection during injections^[12]. Real-world data plays a important role in evaluating the safety and effectiveness of orphan drugs like nusinersen^[11, 13, 14], and gathering real-world safety data from the FDA Adverse Event Reporting System (FAERS) can provide valuable insights.

Researchers can utilize the spontaneous reporting data in FAERS for signal detection and to assess the relationship between drugs and AEs^[15]. It includes AEs data reported by consumers, physicians, pharmacists, and other health-professional following medication use. The database is continuously updated and can be downloaded from the FDA website. Combining with the FDA-approved Prescribing Information (PI)^[16], Our study retrospectively analysed the AEs reported from January 2017 to December 2023 related to nusinersen by FAERS. We identified some AEs not mentioned in the PI, providing valuable information to drug monitoring.

2 Material and methods

2.1 Data source and pre-processing

To prevent duplicate reports from being included, we followed specific criteria for data processing. When CASEIDs are the same, the nearest FDA_DT is selected. If both CASEIDs and FDA_DTs are the same, the higher PRIMARYID is selected^[17, 18]. The data

search covered the period from January 2017 to December 2023. Relevant AEs data was extracted for pre-processing and standardization before being mapped to MedDRA 26.1. Based on the structural hierarchy of MedDRA terminology, discrepant AEs were categorized into preferred terms (PTs) and system organ classes (SOCs). The data extraction steps for this study are illustrated in Fig. 1. Brand names SPINRAZA and common names NUSINERSEN were searched for identification purposes. Following the collection of relevant reports, only the prime suspects (PSs) were retained to enhance accuracy^[19].

2.2 Data mining and statistical analysis

This study utilized the disproportionality analysis method, a commonly employed approach in pharmacovigilance analysis^[20, 21], to detect potential signals between nusinersen and AEs. The reporting odds ratio (ROR), proportional reporting ratio (PRR), Bayesian confidence propagation neural network (BCPNN), and multi-item gamma Poisson shrinker (MGPS) were employed for the analysis. Only when all four signal detection methods are met simultaneously can the signal be accurately identified. The necessary 2x2 contingency table for the algorithm was provided in **Supplement** Table 1, while Table 1 detailed the formulation and criteria for signal detection. A signal is deemed significant if its result exceeds the threshold value, with a higher value indicating a stronger signal. In addition, we described time-to-onset (TTO) of AEs.

Data processing and statistical analyses were performed using SAS (version 9.4), Microsoft excel 2019 and R (version 4.3.2).

Table 1
Detailed formulas for disproportionality analysis.

Algorithms	Equation	Criteria
ROR	$ROR = ad/bc$	$a \geq 3, ROR \geq 3$, lower limit of 95%CI > 1
	$95\%CI = \ln(ROR) \pm 1.96(1/a + 1/b + 1/c + 1/d)^{0.5}$	
PRR	$PRR = [a(c + d)]/[c(a + b)]$	$a \geq 3, PRR \geq 2$, lower limit of 95%CI > 1
	$95\%CI = \ln(ROR) \pm 1.96(1/a - 1/a + b + 1/c - 1/c + d)^{0.5}$	
BCPNN	$IC = \log_2 a(a + b + c + d)/((a + c)(a + b))$	IC025 > 0
	$95\%CI = E(IC) \pm 2V(IC)^{0.5}$	
MGPS	$EBGM = a(a + b + c + d)/(a + c)(a + b)$	EBGM05 > 2
	$95\%CI = \ln(EBGM) \pm 1.96(1/a + 1/b + 1/c + 1/d)^{0.5}$	

IC: information component; IC025: the lower limit of the 95% CI of the IC; E (IC): the IC expectations; V (IC): the variance of IC; EBGM: empirical Bayesian geometric mean; EBGM05: the lower limit of the 95% CI of EBGM.

3 Results

3.1 General characteristics

A total of 5899 AEs reports associated with nusinersen were analyzed from January 2017 to December 2023, as detailed in Table 2. Among these AEs, the highest number of annual reports submitted was 1132 (19.19%) in 2018. 2720 (46.11%) were from females and 2648 (44.89%) were from males. Patients younger than 18 years accounted for 2208 (37.43%), while those older than 65 years contributed 51 (0.86%). The United States had the highest number of reports at 4423 (74.98%). Consumers provided more than 60% of the reports. The most frequently reported serious AE was hospitalization, totaling 1852 (54.53%), followed by other serious events at 824 (24.26%) and death at 615 (18.11%).

Table 2
Demographic characteristics of AEs reported in the FAERS database
(January 2017-December 2023) with nusinersen as the PS drug.

Characteristics	Case number, n	Case proportion, %
Number of events	5899	
Reporting Year		
2017	532	9.02
2018	1132	19.19
2019	966	16.38
2020	753	12.76
2021	936	15.87
2022	813	13.78
2023	767	13.00
Gender		
Female	2720	46.11
Male	2648	44.89
Unkown	531	9.00
Age(years)		
< 18	2208	37.43
18–65	1160	19.67
>=65	51	0.86
unknow	2480	42.04
Reported Countries (top five)		
United States	4423	74.98
Brazil	144	2.44
Turkey	125	2.12
Argentina	102	1.73
Japan	93	1.58
Reporter		
Consumer	3545	60.09
Physician	1648	27.94
Pharmacist	379	6.42
Other health-professional	317	5.37
unkown	10	0.17
Serious Outcomes		
hospitalization	1852	54.53
other serious	824	24.26

Characteristics	Case number, n	Case proportion, %
death	615	18.11
life threatening	79	2.33
disability	24	0.71
congenital anomaly	2	0.06

3.2 Signal detection

The signal intensity and reported cases of nusinersen at the System Organ Class (SOC) level are displayed in Table 3. Nusinersen AEs are observed in 24 different SOC. The study identified several significant SOC categories that satisfied at least one of four algorithms. These included infections and infestations (SOC: 10021881, n = 1912), injury, poisoning and procedural complications (SOC: 10022117, n = 1698), nervous system disorders (SOC: 10029205, n = 1362), respiratory, thoracic and mediastinal disorders (SOC: 10038738, n = 1087), musculoskeletal and connective tissue disorders (SOC: 10028395, n = 1005), investigations (SOC: 10022891, n = 978), and congenital, familial and genetic disorders (SOC: 10010331, n = 112).

A total of 227 Preferred Terms (PTs) met the criteria of all four algorithms simultaneously are described in **Supplementary Table 2**. Headache (ROR = 3.46, n = 437), post lumbar puncture syndrome (ROR = 6144.25, n = 427), protein urine present (ROR = 98.14, n = 98), pyrexia (ROR = 4.4, n = 298), pneumonia (ROR = 4.3, n = 285) and back pain (ROR = 4.37, n = 204) and other PTs were in accordance with the PI. The top 50 significant PTs were extracted based on descending order of ROR values, as presented in Table 4. The leading five PTs included post lumbar puncture syndrome (ROR = 6144.25, n = 427), traumatic lumbar puncture (ROR = 4982.87, n = 24), neuromuscular scoliosis (ROR = 3419.98, n = 3), neurological procedural complication (ROR = 595.49, n = 18) and cerebrospinal fluid (CSF) pressure increased (ROR = 705.13, n = 21). Unexpected AEs were identified in the study, which were not initially reported in the PI. These included traumatic lumbar puncture (ROR = 4982.87, n = 24), neuromuscular scoliosis (ROR = 3419.98, n = 3), CSF pressure increased (ROR = 705.13, n = 21), neurological procedural complication (ROR = 595.49, n = 18), mechanical ventilation complication (ROR = 533.95, n = 11), cerebrospinal fluid leakage (ROR = 164.38, n = 62), CSF protein increased (ROR = 134.2, n = 11), procedural anxiety (ROR = 127.79, n = 14), arachnoid cyst (ROR = 84.45, n = 4), chronic respiratory failure (ROR = 73.21, n = 17) and sudden infant death syndrome (ROR = 68.68, n = 5). Thrombocytopenia was initially proposed in the PI, but our study revealed a contrasting platelet count increase (ROR = 3.5, n = 10). Hypersensitivity reactions (e.g., angioedema, urticaria, and rash) were also mentioned in the PI. However, these reactions did not meet the criteria for at least one of the four disproportionality analysis methods. Additional details on unexpected PTs corresponding to the PI can be found in **Supplementary Table 2**.

We also collected AEs occurring in people aged 65 years that were not mentioned in the PI, and **Supplementary Table 3** revealed five significant PTs: procedural pain (ROR = 113.94, n = 3), road traffic accident (ROR = 68.75, n = 3), sars-cov-2 test positive (ROR = 121.3, n = 4), pain in extremity (ROR = 5.68, n = 3) and headache (ROR = 5.28, n = 4).

Table 3
Signal strength of AEs of nusinersen at the System Organ Class (SOC) level in FAERS .

SOCs	Cases number	ROR(95% CI)	PRR(95% CI)	IC(IC025)	EBGM(EBGM05)
general disorders and administration site conditions	1934	0.81(0.77, 0.85)	0.84(0.81, 0.87)	-0.25(-0.32)	0.84(0.81)
infections and infestations	1912	3(2.86, 3.15)	2.69(2.59, 2.8)	1.43(1.36)	2.69(2.58)
injury, poisoning and procedural complications	1698	1.18(1.12, 1.24)	1.16(1.12, 1.21)	0.21(0.14)	1.16(1.11)
nervous system disorders	1362	1.4(1.32, 1.48)	1.36(1.28, 1.44)	0.44(0.36)	1.36(1.29)
respiratory, thoracic and mediastinal disorders	1087	1.9(1.79, 2.02)	1.82(1.72, 1.93)	0.87(0.78)	1.82(1.73)
musculoskeletal and connective tissue disorders	1005	1.55(1.45, 1.65)	1.51(1.42, 1.6)	0.59(0.5)	1.51(1.43)
investigations	978	1.32(1.23, 1.41)	1.29(1.22, 1.37)	0.37(0.28)	1.29(1.22)
gastrointestinal disorders	804	0.74(0.69, 0.8)	0.76(0.72, 0.81)	-0.4(-0.5)	0.76(0.72)
metabolism and nutrition disorders	272	1.04(0.93, 1.18)	1.04(0.92, 1.17)	0.06(-0.11)	1.04(0.94)
psychiatric disorders	263	0.36(0.31, 0.4)	0.37(0.33, 0.42)	-1.44(-1.61)	0.37(0.33)
cardiac disorders	259	0.96(0.85, 1.09)	0.96(0.85, 1.08)	-0.05(-0.23)	0.96(0.87)
renal and urinary disorders	181	0.67(0.58, 0.78)	0.68(0.59, 0.78)	-0.56(-0.77)	0.68(0.6)
skin and subcutaneous tissue disorders	172	0.22(0.19, 0.25)	0.23(0.2, 0.27)	-2.14(-2.36)	0.23(0.2)
vascular disorders	137	0.54(0.46, 0.64)	0.55(0.47, 0.64)	-0.86(-1.11)	0.55(0.48)
congenital, familial and genetic disorders	112	3.12(2.59, 3.76)	3.1(2.6, 3.7)	1.63(1.36)	3.1(2.65)
blood and lymphatic system disorders	67	0.3(0.24, 0.39)	0.31(0.25, 0.39)	-1.7(-2.04)	0.31(0.25)
eye disorders	56	0.22(0.17, 0.28)	0.22(0.17, 0.28)	-2.17(-2.55)	0.22(0.18)
hepatobiliary disorders	48	0.44(0.34, 0.59)	0.45(0.34, 0.59)	-1.16(-1.57)	0.45(0.35)
immune system disorders	48	0.3(0.22, 0.39)	0.3(0.23, 0.39)	-1.74(-2.14)	0.3(0.24)
ear and labyrinth disorders	47	0.83(0.62, 1.1)	0.83(0.62, 1.11)	-0.27(-0.68)	0.83(0.65)
neoplasms benign, malignant and unspecified (incl cysts and polyps)	47	0.1(0.08, 0.14)	0.11(0.08, 0.15)	-3.22(-3.63)	0.11(0.08)
reproductive system and breast disorders	20	0.22(0.14, 0.34)	0.22(0.14, 0.34)	-2.17(-2.79)	0.22(0.15)
endocrine disorders	13	0.38(0.22, 0.54)	0.38(0.22, 0.54)	-1.39(-2.15)	0.38(0.24)

SOCs	Cases number	ROR(95% CI)	PRR(95% CI)	IC(IC025)	EBGM(EBGM05)
		0.65)	0.66)		
pregnancy, puerperium and perinatal conditions	7	0.14(0.07, 0.29)	0.14(0.07, 0.29)	-2.85(-3.85)	0.14(0.07)

Values in red and bold indicates significant signals in algorithms.

Table 4
Signal strength of AEs of nusinersen at the top 50(ROR) Preferred terms (PTs) level in FAERS.

SOCs	PTs	Case Reports	ROR(95% CI)	PRR(95% CI)	IC(IC025)	EBGM(EBGM05)
injury, poisoning and procedural complications	post lumbar puncture syndrome	427	6144.25(5127.3, 7362.91)	5934.89(4975.12, 7079.82)	10.69(10.5)	1647.63(1416.15)
injury, poisoning and procedural complications	traumatic lumbar puncture	24	4982.87(2440.34, 10174.41)	4973.33(2455.89, 10071.3)	10.61(9.87)	1563.73(860.5)
musculoskeletal and connective tissue disorders	neuromuscular scoliosis	3	3419.98(571.39, 20469.74)	3419.16(574.53, 20348.25)	10.42(8.59)	1368.26(306.16)
investigations	csf pressure increased	21	705.13(432.14, 1150.56)	703.95(431.26, 1149.07)	9.07(8.4)	538.08(357.21)
injury, poisoning and procedural complications	neurological procedural complication	18	595.49(354.37, 1000.67)	594.64(357.22, 989.85)	8.88(8.17)	471.82(305.6)
investigations	csf red blood cell count positive	4	536.51(180.5, 1594.69)	536.34(178.96, 1607.39)	8.76(7.36)	434.37(174.58)
injury, poisoning and procedural complications	mechanical ventilation complication	11	533.95(276.88, 1029.71)	533.49(279.4, 1018.66)	8.76(7.86)	432.5(249.65)
injury, poisoning and procedural complications	procedural vomiting	34	268.9(188.39, 383.8)	268.17(188.45, 381.62)	7.91(7.4)	240.05(178.24)
injury, poisoning and procedural complications	musculoskeletal procedural complication	4	268.25(95.18, 756.07)	268.17(94.9, 757.8)	7.91(6.56)	240.05(100.87)
injury, poisoning and procedural complications	respiratory tract procedural complication	4	268.25(95.18, 756.07)	268.17(94.9, 757.8)	7.91(6.56)	240.05(100.87)
injury, poisoning and procedural complications	procedural nausea	35	239.53(169.02, 339.46)	238.86(167.85, 339.91)	7.76(7.26)	216.3(161.57)
injury, poisoning and procedural complications	tracheostomy malfunction	9	228.11(114.94, 452.71)	227.94(114.79, 452.63)	7.7(6.76)	207.31(116.83)
investigations	physical examination abnormal	9	211.65(106.88, 419.1)	211.49(106.5, 419.97)	7.6(6.66)	193.62(109.32)
investigations	csf white blood cell count increased	4	168.9(61.16, 466.43)	168.85(60.94, 467.88)	7.3(5.98)	157.27(67.22)
nervous system disorders	cerebrospinal fluid leakage	62	164.38(126.96, 212.83)	163.57(126.78, 211.04)	7.25(6.89)	152.69(123.01)
injury, poisoning and procedural complications	procedural headache	13	156.95(89.45, 275.39)	156.79(88.81, 276.8)	7.2(6.42)	146.76(91.68)
musculoskeletal and connective tissue disorders	scoliosis	167	143.25(122.4, 167.65)	141.35(120.84, 165.35)	7.06(6.83)	133.16(116.74)
investigations	csf protein increased	11	134.2(73.04, 246.58)	134.08(73.03, 246.17)	6.99(6.14)	126.69(76.15)

SOCs	PTs	Case Reports	ROR(95% CI)	PRR(95% CI)	IC(IC025)	EBGM(EBGM05)
investigations	urine protein/creatinine ratio abnormal	5	134.14(54.42, 330.63)	134.08(54.43, 330.31)	6.99(5.79)	126.69(59.56)
injury, poisoning and procedural complications	meningitis chemical	5	132.58(53.8, 326.69)	132.53(53.8, 326.49)	6.97(5.78)	125.3(58.92)
investigations	neurological examination abnormal	10	131.86(69.69, 249.51)	131.76(69.01, 251.59)	6.96(6.08)	124.61(73.09)
nervous system disorders	intracranial hypotension	5	129.56(52.61, 319.08)	129.51(52.57, 319.05)	6.94(5.75)	122.6(57.68)
injury, poisoning and procedural complications	post procedural fever	9	128.31(65.55, 251.17)	128.22(65.85, 249.67)	6.92(6)	121.44(69.23)
psychiatric disorders	procedural anxiety	14	127.79(74.58, 218.98)	127.65(75.2, 216.7)	6.92(6.17)	120.93(77.06)
injury, poisoning and procedural complications	post procedural contusion	10	120.7(63.88, 228.05)	120.61(64.42, 225.83)	6.84(5.96)	114.6(67.29)
infections and infestations	enterovirus infection	23	107.19(70.53, 162.9)	106.99(70.89, 161.47)	6.68(6.08)	102.24(72.03)
investigations	csf volume increased	3	103.64(32.58, 329.66)	103.61(32.6, 329.32)	6.63(5.18)	99.15(37.65)
investigations	protein urine present	98	98.14(80.11, 120.22)	97.38(80.05, 118.47)	6.55(6.25)	93.43(78.84)
musculoskeletal and connective tissue disorders	kyphoscoliosis	3	97.71(30.76, 310.37)	97.69(30.73, 310.5)	6.55(5.1)	93.72(35.63)
infections and infestations	rhinovirus infection	89	96.46(77.97, 119.34)	95.78(77.2, 118.83)	6.52(6.22)	91.96(76.96)
nervous system disorders	arachnoid cyst	4	84.45(31.13, 229.13)	84.42(31.07, 229.39)	6.35(5.06)	81.44(35.33)
injury, poisoning and procedural complications	procedural dizziness	5	75.01(30.77, 182.85)	74.98(31.04, 181.13)	6.18(5.01)	72.63(34.46)
respiratory, thoracic and mediastinal disorders	chronic respiratory failure	17	73.21(45.16, 118.7)	73.11(44.79, 119.34)	6.15(5.47)	70.87(47.3)
respiratory, thoracic and mediastinal disorders	bronchial secretion retention	23	72.95(48.14, 110.53)	72.82(48.25, 109.9)	6.14(5.55)	70.59(49.86)
nervous system disorders	bulbar palsy	3	69.8(22.12, 220.19)	69.78(21.95, 221.79)	6.08(4.64)	67.74(25.9)
infections and infestations	pneumonia respiratory syncytial viral	12	69.49(39.12, 123.45)	69.42(39.32, 122.56)	6.07(5.28)	67.4(41.67)
general disorders and administration site conditions	sudden infant death syndrome	5	68.68(28.21, 167.23)	68.66(28.42, 165.86)	6.06(4.88)	66.68(31.67)

SOCs	PTs	Case Reports	ROR(95% CI)	PRR(95% CI)	IC(IC025)	EBGM(EBGM05)
infections and infestations	human bocavirus infection	4	62.9(23.29, 169.89)	62.88(23.14, 170.86)	5.94(4.65)	61.22(26.66)
respiratory, thoracic and mediastinal disorders	respiratory muscle weakness	4	62.9(23.29, 169.89)	62.88(23.14, 170.86)	5.94(4.65)	61.22(26.66)
investigations	csf test abnormal	4	59.61(22.09, 160.89)	59.59(21.93, 161.92)	5.86(4.58)	58.1(25.31)
injury, poisoning and procedural complications	airway complication of anaesthesia	3	57.97(18.43, 182.34)	57.95(18.59, 180.62)	5.82(4.38)	56.54(21.67)
infections and infestations	respiratory syncytial virus bronchiolitis	6	57.01(25.36, 128.2)	56.99(25.52, 127.29)	5.8(4.71)	55.62(28.23)
injury, poisoning and procedural complications	procedural pain	281	56.56(50.19, 63.75)	55.32(49.18, 62.22)	5.76(5.58)	54.03(48.88)
respiratory, thoracic and mediastinal disorders	increased bronchial secretion	23	53.43(35.32, 80.82)	53.33(35.34, 80.49)	5.7(5.12)	52.14(36.88)
investigations	urine protein/creatinine ratio increased	10	53.3(28.47, 99.8)	53.26(28.45, 99.72)	5.7(4.84)	52.06(30.8)
gastrointestinal disorders	cyclic vomiting syndrome	3	51.43(16.37, 161.52)	51.42(16.5, 160.26)	5.65(4.22)	50.3(19.31)
congenital, familial and genetic disorders	developmental hip dysplasia	10	50.25(26.85, 94.05)	50.21(26.82, 94.01)	5.62(4.76)	49.15(29.09)
infections and infestations	respiratory tract infection viral	26	49.7(33.68, 73.32)	49.59(33.51, 73.39)	5.6(5.05)	48.56(35.07)
nervous system disorders	arachnoiditis	5	49.36(20.35, 119.73)	49.34(20.42, 119.19)	5.59(4.43)	48.31(23.02)
musculoskeletal and connective tissue disorders	joint contracture	7	48.09(22.74, 101.68)	48.06(22.82, 101.22)	5.56(4.55)	47.09(25.17)

3.3 TTO description

We extracted the TTO of nusinersen AEs from the FAERS. After removing any missing or incorrect TTO reports, a total of 1884 AEs with available TTO were included. The median TTO was 119.5 days and the interquartile range (IQR) of 14–492 days. Regarding the distribution of AEs over time, Fig. 2 illustrates the TTO and the corresponding percentage of those reports. Our data showed that there were 373 AEs less than 7 days (19.8%) and still had 603 AEs greater than one year (32.01%).

4 Discussion

Efficacy and safety data for drugs are typically obtained from preclinical and clinical trials^[22]. However, limitations such as trial design and small sample sizes may lead to inadequate descriptions of nusinersen safety profile. We conducted a pharmacovigilance analysis using real-world data, which allowed us to identify potential AEs and provide insights into the safety profile of nusinersen in clinical practice.

The majority of AEs in this study were observed in patients under 18 years of age, with only a small percentage occurring in patients older than 65 years. Our findings indicated that nusinersen had the highest reported incidence of AEs in 2018, with a subsequent slowing in the rate of increase. This trend could possibly be attributed to the introduction of competing treatments such as risdiplam (small molecule therapeutics) and onasemnogene abeparvovec (gene therapy drugs).

Treatment of complex neurological disorders often involves intrathecal administration through lumbar puncture^[23]. Nusinersen also utilizes intrathecal administration to penetrate the blood-brain barrier and target the central nervous system. While studies have demonstrated that intrathecal administration of nusinersen is tolerated^[24], it remains a significant challenge in patients with severe scoliosis and spinal deformities^[25]. The PI recommended considering ultrasound or other imaging techniques for younger patients and those with scoliosis to assist in guiding intrathecal administration. Notable AEs such as traumatic lumbar puncture, neurological procedural complication, arachnoid cyst and CSF protein increased were not directly attributed to nusinersen, but rather to the intrathecal administration procedure. The PI has recommended removing an equal volume of CSF as the injected drug before drug delivery. It is hypothesized that changes in CSF volume are associated with CSF pressure increase and cerebrospinal fluid leakage. Following nusinersen treatment, physicians should monitor for symptoms indicative of post-lumbar puncture syndrome and assess intracranial pressure^[26]. Due to the need for multiple administrations over an extended period, nusinersen treatment through repeated lumbar punctures can lead to various complications, including the risk of infusion site infection or pain. Moreover, it is considered less cost-effective compared to other drugs for treating SMA^[27, 28]. Addressing this complex issue requires collaboration between neurologists and surgeons to develop convenient, safe, and efficient methods for multiple drug deliveries.

Respiratory complications are a common occurrence in SMA and significantly contribute to morbidity and mortality in affected patients^[29]. Additionally, orthopedic issues are frequently observed in children with SMA^[1]. While nusinersen has shown effectiveness in enhancing motor milestones and function, it did not provide a cure for symptomatic patients^[25], and can not stop the progression of scoliosis^[30]. It is important to note that neuromuscular scoliosis, mechanical ventilation complications, and chronic respiratory failure are associated with the disease process that necessitate diligent nursing care to prevent or delay the development of associated AEs.

Procedural anxiety may be linked to decreased patient adherence due to repeated intrathecal administration^[11]. Compliance is a crucial factor in the long-term use of nusinersen. Studies have shown that discontinuation and non-adherence are more common in nusinersen therapy, leading to higher rates of comorbidities, increased healthcare resource utilization, and elevated patient costs^[31, 32]. Poor compliance can result in delayed or interrupted SMA treatment, leading to disease progression and higher treatment expenses. Therefore, it is essential to investigate the impact of compliance on nusinersen treatment.

The PI noted the lack of research data for individuals over 65 years old. We found there were five significant PTs in aged 65 and over, but pain in extremity was insignificant in patients 18–65 years of age. Further research is needed on whether AEs in patients 65 years and older will differ from those in younger patients.

Our findings revealed that the distribution of the TTO was extensive, with AEs persisting in 32.01% of cases even after 1 year of treatment. The loading dose of nusinersen requires a minimum of 63 days for administration, followed by regular maintenance doses every four months^[16, 33]. Despite the treatment, there remains a risk associated with each intrathecal administration. It is crucial to monitor the progression of SMA and address orthopedic and respiratory issues that may arise. Vigilance is necessary to promptly manage any AEs that may arise.

This study acknowledges several limitations. Firstly, spontaneous reporting is known to suffer from underreporting, incomplete content, and selective reporting. Secondly, there is a significant lack of detailed data for 42.04% of the age group, and 60.09% of the reporters are consumers, potentially introducing bias. Additionally, the study is unable to determine the incidence of AEs due to the unknown total number of people treated and the absence of a control group. Finally, the disproportionate analyses conducted could not establish causality. Despite these limitations, the findings of this study may contribute valuable safety data for future clinical studies involving nusinersen.

5 Conclusion

This study conducted pharmacovigilance analyses of nusinersen in FAERS, highlighting the potential risks associated with nusinersen treatment and the TTO of AEs. Physicians should remain vigilant towards these potential AEs and prioritize attentive care for their patients. The findings of our study could contribute valuable safety data on nusinersen, serving as a foundation for future research endeavors.

Declarations

6 Conflict of Interest

The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

7 Data availability statement

The original contributions proposed in this study are included in the article/supplementary materials. For further inquiries, please contact the corresponding author directly.

8 Ethics statement

The data in the FAERS database does not raise ethical concerns.

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Figures

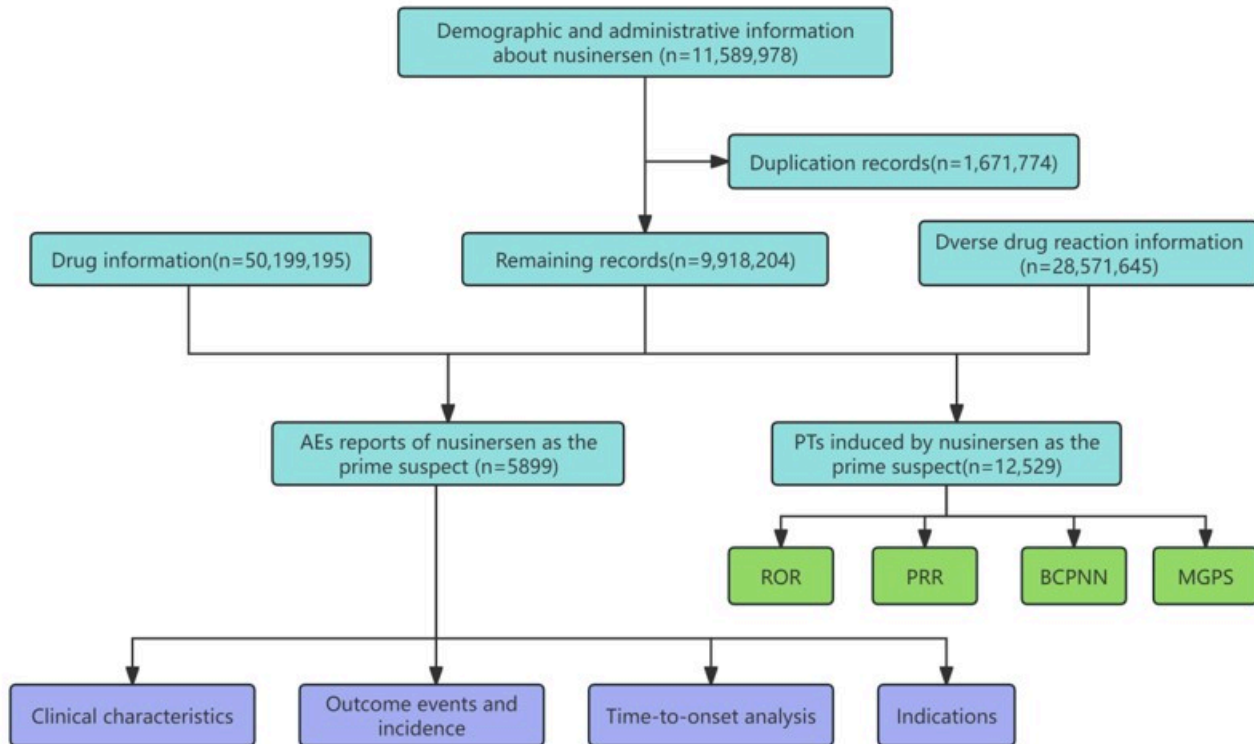


Figure 1

The flow diagram of selecting nusinersen-related AEs from FAERS.

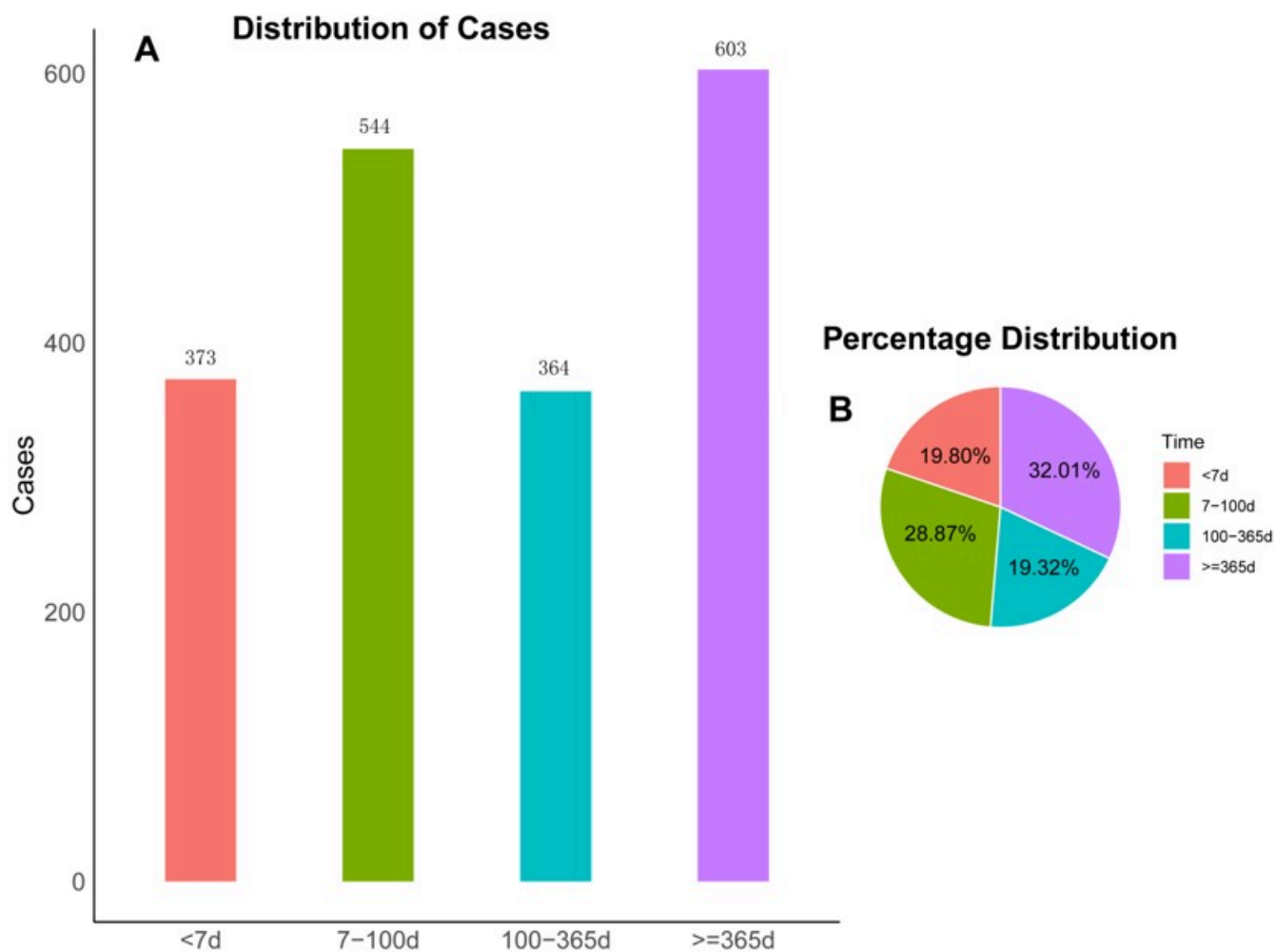


Figure 2

TTO of nusinersen AEs.

Supplementary Files

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- [SupplementTable13.pdf](#)